

Springer Texts in Statistics

Gareth James
Daniela Witten
Trevor Hastie
Robert Tibshirani

An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Week 12-13

Instructor:

Jeffrey Zheng

*Examples Inspired By
CAS, SOA, Coaching
Actuaries ©*

Key Formulas This Week

- GLM: $\mathbf{x}_i' \boldsymbol{\beta} = g(\mu_i)$, $E[y] = b'(\theta)$, $Var(y) = \varphi b''(\theta) = \varphi v(\mu_i)$
- Linear Exponential: $f(y; \theta, \varphi) = e^{\left[\frac{y\theta - b(\theta)}{\varphi} + S(y, \varphi) \right]}$,
- Overdispersion: $Var(y_i) = c\varphi \frac{b''(\theta_i)}{w_i}$
- Normal: $\mu = \theta$, $v(\mu) = 1$, Identity
- Bernoulli: $\mu = \frac{e^\theta}{1+e^\theta}$, $v(\mu) = \mu(1 - \mu)$, Logit
- Poisson: $\mu = e^\theta$, $v(\mu) = \mu$, Log
- Tweedie: $v(\mu) = \mu^p$ for $1 < p < 2$
- Gamma: $\mu = -\frac{1}{\theta}$, $v(\mu) = \mu^2$, Negative Inverse
- Inverse Gaussian: $\mu = (-2\theta)^{-\frac{1}{2}}$, $v(\mu) = \mu^3$, Negative Squared Inverse

Key Formulas This Week 2

- Deviance: $D = 2(l_{SAT} - l_{\hat{\theta}})$
- $e_{i,Pearson} = \frac{y_i - \hat{\mu}_i}{\sqrt{Var(y_i)}}$
- $d_{i,Deviance} = \text{sign}(y_i - \hat{\mu}_i)\sqrt{D}$
- $R_{maxscaled}^2 = \frac{1 - e^{\frac{2(l_0 - l)}{n}}}{1 - e^{\frac{2l_0}{n}}}$, $R_{pseudo}^2 = \frac{l_{\hat{\theta}} - l_0}{l_{SAT} - l_0}$
- $LRT = 2(l_a - l_0) = D_0 - D_a \sim \chi_q^2$

GLMs

- **Generalized linear models** extend the linear model by relaxing assumptions.
- GLMs retain the feature that the mean of the response is a function of a _____
- Notation: $\mu_i = E[y_i]$
- $\eta_i = \mathbf{x}_i' \boldsymbol{\beta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$

Linear Model Assumptions

- Multiple linear regression model assumptions:
- 1. $y_i = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \varepsilon_i$ (GLM relaxes)
- 2. x_i' s are **non-stochastic variables** (GLM assumption)
- 3. $E[\varepsilon_i] = 0$ (GLM assumption)
- 4. Homoscedasticity: $Var(y_i) = \sigma^2$ (GLM relaxes)
- 5. y_i' s / ε_i' s are **independent** (GLM assumption)
- 6. y_i' s / ε_i' s are **normally distributed** (GLM relaxes)
- 7. Predictors are **non-collinear** (GLM assumption)

Relaxing Form (#1)

- For normal linear regression,
- $E[y_i] = \mu_i = \mathbf{x}_i' \boldsymbol{\beta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$
 - μ_i : Equivalent notation for $E[y_i]$
- In GLMs, the left-hand can be any _____

Link Function (#1)

- $g(\mu) = x_i' \boldsymbol{\beta} = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$ is the **systematic component** of the model and it is set equal to $g(\mu_i)$.
- The function $g(\cdot)$ is the **link function** that relates the systematic component to μ_i . It allows $x_i' \boldsymbol{\beta}$ to take on _____

Relaxing Form (#1) 2

Link function Linear predictor

$$\ln \lambda_i = b_0 + b_1 x_i$$

$$y_i \sim \text{Poisson}(\lambda_i)$$

Probability distribution

- For example: $\ln \frac{\mu_i}{1-\mu_i} = \mathbf{x}_i' \boldsymbol{\beta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$ uses the logit link function
- $\ln \mu_i = \mathbf{x}_i' \boldsymbol{\beta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$ uses the log link function

Canonical Link Function

- The canonical link function for a distribution is $g(\mu_i)$ such that it equals θ_i (from the linear exponential decomposition).
- Normal Distribution: $g(\mu_i) = \mu_i$ (Identity)
- Bernoulli: $g(\mu_i) = \text{logit}(\mu_i) = \ln \frac{\mu_i}{1-\mu_i}$ (Logit)
- Poisson: $g(\mu_i) = \ln \mu_i$ (Log)
- Gamma $f(x) = \frac{\gamma^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\gamma x}$: $g(\mu_i) = -\frac{1}{\mu_i}$ (Negative Inverse)
- Inverse Gaussian $f(x) = \sqrt{\frac{\beta}{2\pi x^3}} e^{\frac{-\beta(x-\alpha)^2}{2\alpha^2 x}}$: $g(\mu_i) = \frac{-1}{2\mu_i^2}$

Link Function Convention

- Link functions are scale equivalent, meaning negative signs/multiplicative constants are convention-dependent.
- ie. Using Gamma: $g(\mu_i) = -\frac{1}{\mu_i}$ vs. $g(\mu_i) = \frac{1}{\mu_i}$ will yield differently parametrized models, but their predictions will be identical as they are scale equivalent link functions

Relaxing Variance (#3)

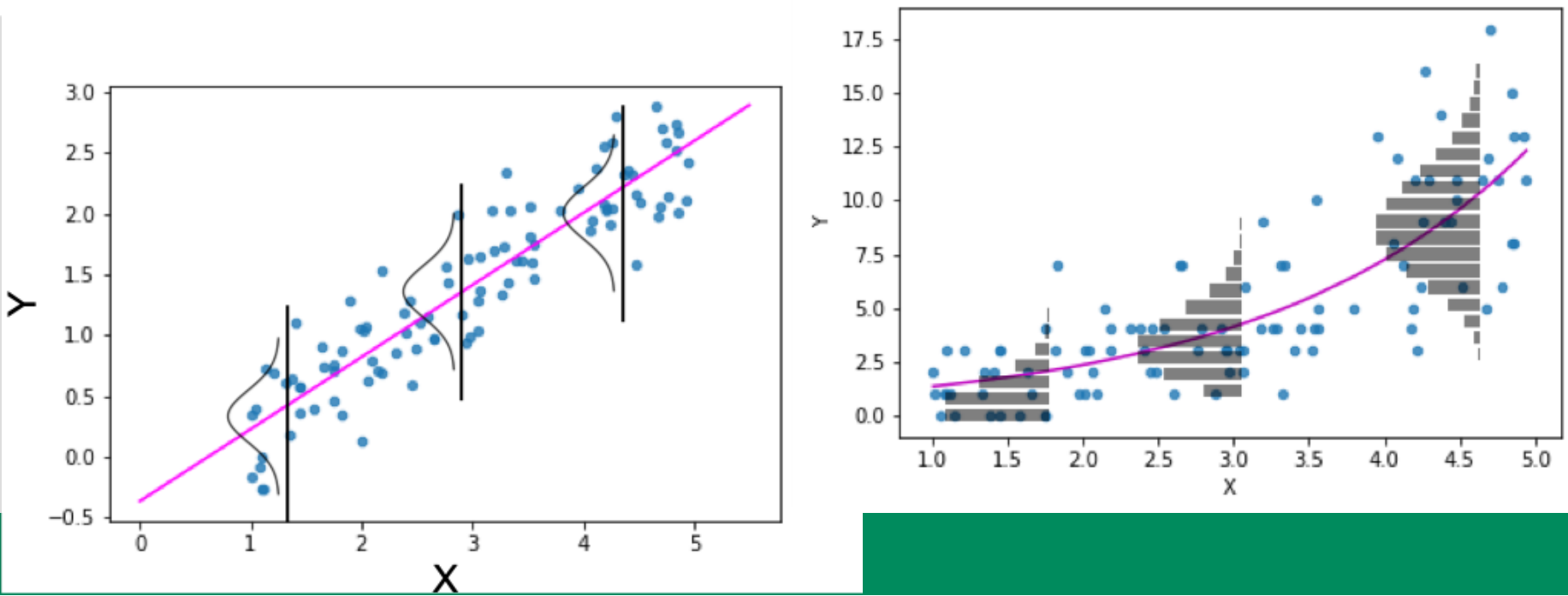
- The variance function $v(\mu)$ in a GLM relates the variance of the GLM to its mean, no longer requiring _____
- $Var(y) =$
 - φ : scale parameter (a constant)

Relaxing Variance (#3) 2

- Normal Distribution: $v(\mu) = 1$ so homoscedastic
- Common Relations:
- Bernoulli/Logistic GLM: $v(\mu) = \mu(1 - \mu)$
- Poisson GLM: $v(\mu) = \mu$
- Gamma GLM: $v(\mu) = \mu^2$

Relaxing Variance Visual

- <https://towardsdatascience.com/generalized-linear-models-9cbf848bb8ab>
- Linear model (left) vs. Poisson GLM (right)



Score Equations

- GLM coefficients are estimated using maximum likelihood estimation, which _____
- In simple linear equation $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ for example, the score equations are:
 - $\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$ (residuals sum to 0)
 - $\sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0$

GLM Predictions

- Once the GLM parameters are estimated, their primary use is to estimate the mean of the response variable, given a vector of explanatory values $\mathbf{x}$.
- These always involve:
- Estimating $g(\hat{\mu}) = \mathbf{x}'\hat{\boldsymbol{\beta}} = \hat{\beta}_0 + \hat{\beta}_1x_1 + \cdots + \hat{\beta}_px_p$
- Inverting g to obtain $\hat{\mu}$.
- As a MLE, $\hat{\mu}$ is consistent, asymptotically unbiased, and asymptotically normal.

Concept Check #1

- Determine which statements are true.
- I. The logit link function pairs well with an exponentially distributed response variable.
- II. Multiple linear regression is a special case of a GLM.
- III. The canonical link for binomial is $N^{-1}(\mu)$.
- IV. The canonical link for gamma is $\frac{-1}{\mu}$.

Example 1 (CAS)

- 3 separate GLMs are fit using the model:
- $g(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$
- Each model uses its canonical link function.
- Model I: Gamma (estimated $\varphi = 1.5$)
- Model II: Poisson
- Model III: Bernoulli
- Assume all 3 models have the same parameter estimates: $\hat{\beta}_0 = 2, \hat{\beta}_1 = 1, \hat{\beta}_2 = -1$.
- Determine the predicted values and variances of the predicted means at $X_1 = 5, X_2 = 2$

Example 1 Sol

- $g(\hat{\mu}) = \mathbf{x}'\hat{\boldsymbol{\beta}} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$
- $Var(y) = \varphi v(\mu)$
- Bernoulli: $\mu = \frac{e^\theta}{1+e^\theta}$, $v(\mu) = \mu(1 - \mu) = Var(Y)$, $g(\mu) = \ln \frac{\mu}{1-\mu}$ (logit)
- Poisson: $\mu = e^\theta$, $v(\mu) = \mu = Var(Y)$, $g(\mu) = \ln \mu$ (log)
- Gamma: $\mu = -\frac{1}{\theta}$, $v(\mu) = \mu^2$, $Var(Y) = \varphi \mu^2$, $g(\mu) = -\frac{1}{\mu}$ (inverse)

Example 2 (CAS)

- A GLM is used to estimate expected loss by smoking status X_1 and gender X_2 . Non-smoking is the base Smoking level. Male is the base Gender level. Interaction terms are included in the model.

$$\bullet X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \hat{Y} = \begin{bmatrix} 500 \\ 900 \\ 300 \\ 700 \end{bmatrix}$$

- Find the model equation assuming the:
 - A. Identity link function
 - B. Log link function

Example 2 Sol

- $g(\hat{\mu}) = \mathbf{x}'\hat{\boldsymbol{\beta}} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$
- Testable parameter estimation Qs are prediction Qs in disguise.

Relaxing y (#5)

- In GLMs, y_i' s can be from any member of the **linear exponential family** of distributions (of which normal distribution is a part of).
- Normal, Binomial, Poisson, Gamma, Inverse Gaussian are all part of the linear exponential family. The distribution chosen should have a domain compatible with the response.
- Uniform is not in the linear exponential family.

Linear Exponential (#5)

- The distribution of the linear exponential family is:
- $f(y; \theta, \varphi) =$
 - y : Variable in the distribution
 - θ : parameter of interest & is a function of the mean of the response variable
 - φ : Scale/dispersion parameter
 - $b(\theta)$ only depends on θ
 - $S(y, \varphi)$ does not depend on θ

Linear Exponential (#5) 2

- For any member of the linear exponential family,
- Mean Function: $E[y] =$
- Variance Function: $Var(y) =$
- GLM = Response Distribution (Linear Exponential Family) + Chosen Link Function

Scale Parameter

- The scale parameter φ in the linear exponential family is 1 for the Binomial, Poisson, and Exponential distributions.
- It would need to be estimated/provided for other more complex distributions.

Variance Function

- In GLMs, the scale parameter and variance function can vary by observation within the same distribution.
- Let $\varphi_i = \frac{\varphi}{w_i}$
 - φ : Constant (Dispersion parameter)
 - w_i : Known weights by observation
- $Var(y_i) = \varphi_i b''(\theta_i)$

Variance Function 2

- In GLMs, the scale parameter and variance function can vary by observation within the same distribution.
- Let $\varphi_i = \frac{\varphi}{w_i}$
 - φ : Constant (Dispersion parameter)
 - w_i : Known weights by observation
- $Var(y_i) = \varphi_i b''(\theta_i)$

Common Decompositions

Distribution	θ	φ	$b(\theta)$
Normal	μ	σ^2	$\frac{\theta^2}{2}$
Bernoulli	$\ln \frac{\pi}{1 - \pi}$	1	$\ln(1 + e^\theta)$
Poisson	$\ln \lambda$	1	e^θ
Negative Binomial	$\ln(1 - p)$	1	$-r \ln(1 - e^\theta)$
Gamma	$-\frac{\gamma}{\alpha}$	$\frac{1}{\alpha}$	$-\ln(-\theta)$
Inverse Gaussian	$-\frac{1}{2\mu^2}$	$\frac{1}{\lambda}$	$-\sqrt{-2\theta}$

1. Linear Exp. Illustration

- Y is a random variable in the linear exponential family.
- $f(y; \beta) = e^{\left[-\frac{y}{\beta} - 4 \ln \beta + 3 \ln y - \ln 2\right]}, y > 0$
- A. Perform the linear exponential decomposition for the distribution.
- B. Find the variance of Y in terms of β .

1. Linear Exp. Illustration 2

- $f(y; \theta, \varphi) = e^{\left[\frac{y\theta - b(\theta)}{\varphi} + s(y, \varphi) \right]}$
- $Var(Y) = \varphi * b''(\theta)$

Example 3

- Y is a random variable in the linear exponential family.
- $f(y; \theta, \varphi) = e^{\left[\frac{y\theta - b(\theta)}{\varphi} + S(y, \varphi)\right]}$
- $b(\theta) = -\sqrt{-2\theta}$
- $\theta = -0.6$
- The scale parameter is 1.4.
- Calculate $Var(Y)$.

Example 3 Sol

- $E[Y] = b'(\theta)$
- $Var(Y) = \varphi b''(\theta)$

Example 4 (CAS)

- Perform the linear exponential family decomposition for the Poisson distribution.
- $p(y; \lambda) = e^{-\lambda} \frac{\lambda^y}{y!}$
- $f(y; \theta, \varphi) = e^{\left[\frac{y\theta - b(\theta)}{\varphi} + s(y, \varphi) \right]}$
- Hint: Take the natural log, rearrange, and then exponentiate.

Example 4 Sol

- $f(y; \theta, \varphi) = e^{\left[\frac{y\theta - b(\theta)}{\varphi} + S(y, \varphi)\right]}$

Distributions Summary

- Mean, Variance Functions & Canonical Link:
- $E[y] = \mu = b'(\theta), Var(y) = \phi v(\mu)$
- Normal: $\mu = \theta, v(\mu) = 1, Var(Y) = \sigma^2, g(\mu) = \mu$ (identity)
- Bernoulli: $\mu = \frac{e^\theta}{1+e^\theta}, v(\mu) = \mu(1 - \mu) = Var(Y), g(\mu) = \ln \frac{\mu}{1-\mu}$ (logit)
- Poisson: $\mu = e^\theta, v(\mu) = \mu = Var(Y), g(\mu) = \ln \mu$ (log)
- Gamma: $\mu = \frac{1}{\theta}, v(\mu) = \mu^2, Var(Y) = \phi \mu^2, g(\mu) = -\frac{1}{\mu}$ (inverse)
- Inverse Gaussian: $\mu = (-2\theta)^{-\frac{1}{2}}, v(\mu) = \phi \mu^3 = Var(Y), g(\mu) = -\frac{1}{2\mu^2}$ (squared inverse)

Log Likelihood

- The log-likelihood function quantifies how probable a set of statistical parameters values are, given a set of observations.
- $L(\boldsymbol{\beta})$ is the product of the density functions of each individual observation.
- For example: If $f(x) = \beta e^{-\beta x}, x > 0$ and observations of $x_i = \{3, 11, 2\}$ are observed:
- $L(\boldsymbol{\beta}) =$

Log Likelihood 2

- Log-likelihood function: $l(\boldsymbol{\beta}) = \ln L(\boldsymbol{\beta}) = \sum_{i=1}^n \left(\frac{y_i \theta_i - b(\theta_i)}{\varphi_i} + S(y_i, \varphi_i) \right)$
- Continuing the example:
- $l(\boldsymbol{\beta}) =$

Terminology

- **Saturated Model:** 1 parameter for every ____
 - l_{SAT} : log-likelihood for the saturated model
- **Null Model:** Model with only an ____
 - l_0 : log-likelihood for the null model
- $l_0 \leq l(\mathbf{b}) \leq l_{sat}$
 - $l(\mathbf{b})$: log-likelihood from maximum likelihood estimated $\mathbf{b}$'s

Deviance

- **Scaled deviance** compares a model to the likelihood statistic of the **saturated** model, where every observation is its own parameter.
- $D =$
 - l_{SAT} : Log-likelihood value of saturated model
 - $l_{\hat{\theta}}$: Log-likelihood value of proposed model
- Larger deviance indicates _____

Deviance Function Steps

- To calculate scaled deviance, write the log-likelihood function in terms of $\mu_1, \mu_2, \dots, \mu_n$.
- Let $\mu_i = y_i$ in the saturated model, and $\mu_i = \hat{\mu}_i$ in the fitted model.
- D will be in terms of y_i' s and $\hat{\mu}_i'$ s.

Common Deviances

- Normal: $D(\hat{\boldsymbol{\mu}}) = \frac{\sum_i (y_i - \hat{\mu}_i)^2}{\sigma^2} = \frac{SSE}{\sigma^2}$
- Bernoulli: $D(\hat{\boldsymbol{\pi}}) = 2 \sum_i \left[y_i \ln \frac{y_i}{\hat{\pi}_i} + (1 - y_i) \ln \frac{1 - y_i}{1 - \hat{\pi}_i} \right]$
- Poisson: $D(\hat{\boldsymbol{\mu}}) = 2 \sum_i \left[y_i \ln \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right]$
- Exponential: $D(\hat{\boldsymbol{\mu}}) = 2 \sum_i \left[\frac{y_i}{\hat{\mu}_i} + \ln \frac{\hat{\mu}_i}{y_i} - 1 \right]$
- Gamma: $D(\hat{\boldsymbol{\mu}}) = 2\alpha \sum_i \left[\frac{y_i}{\hat{\mu}_i} + \ln \frac{\hat{\mu}_i}{y_i} - 1 \right]$

2. Deviance Illustration

- Derive the deviance formula for a GLM with a normal response for each observation, but common unknown variance σ^2 .

- $$f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

2. Deviance Illustration 2

- To calculate deviance, write the log-likelihood function in terms of $\mu_1, \mu_2, \dots, \mu_n$.
- Let $\mu_i = y_i$ in the saturated model, and $\mu_i = \hat{\mu}_i$ in the fitted model.
- $D = 2(l_{SAT} - l_{\hat{\theta}})$

Example 5 (CAS)

- You are given $y_1, y_2, \dots, y_n$ independent Poisson distributed random variables with respective means μ_i for $i = 1, 2, \dots, n$
- A Poisson GLM was fitted to the data with a log-link function expressed as:
 - $E[y_i] = e^{\beta_0 + \beta_1 x_i}$
- Calculate the observed deviance for testing the adequacy of the model.

x_i	y_i	$\hat{\mu}_i$	$y_i \ln \left(\frac{y_i}{\hat{\mu}_i} \right)$
0	5	6	-0.912
1	9	5	5.29
1	2	5	-1.833
0	3	6	-2.079
0	10	6	5.108
2	8	7	1.068
2	5	7	-1.682

Example 5 Sol

- Poisson: $D(\hat{\mu}) = 2 \sum_i \left[y_i \ln \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right]$

Alternative R^2

- Max Scaled- $R^2 =$
 - l : Log-likelihood of GLM model
 - l_{SAT} : Log-likelihood of saturated model
 - l_0 : Log-likelihood of null model
- Pseudo- $R^2 =$

Concept Check #2

- Determine which statements are true.
- I. A pseudo R^2 of 1 implies a max-scaled R^2 of 1.
- II. A pseudo R^2 of 0 implies a max-scaled R^2 of 0.
- III. A small deviance indicates a poor fit for a model.
- IV. A saturated model has a deviance of zero.
- V. Deviance for normal distributions is proportional to the residual sum of squares.
- VI. Deviance is defined as a measure of distance between saturated and fitted model.

3. Psuedo- R^2 Illustration

- 3 different GLMs using the same distribution and link function are fit to the same dataset with 60 observations.
- Calculate the pseudo- R^2 of the Model B.

Model	# of Predictors	Scaled Deviance
A	0	200
B	5	120
C	59	0

3. Pseudo- R^2 Illustration 2

- $\text{Pseudo-}R^2 = \frac{l - l_0}{l_{SAT} - l_0}$
- $D = 2(l_{SAT} - l_{\hat{\theta}})$

Residuals

- Raw Residuals = $y_i - \hat{\mu}_i$
- *Raw residuals can still help find:*
- Additional variables, remaining non-linear patterns
- Poorly fitting observations (outliers)
- Effect of individual observations on parameters (influential points)
- Time trends (violates GLM)

Raw Residuals Limits

- But, in some non-linear contexts and distributions (ie. binary), raw residuals are unhelpful.
- Appropriate residual behavior depends on _____

Pearson Residuals

- Pearson residuals $e_i =$
- Pearson chi-square statistic =

Over/Under-dispersion

- For distributions where the variance is determined by the mean, **over/under-dispersion** can adjust the variance.
- $c > 1$: Over-dispersion increases variance
- $0 < c < 1$: Under-dispersion reduces variance (Rare)
- $Var(y_i) =$
 - c : Original distribution scale parameter
- $\hat{\varphi} = \frac{\sum_{i=1}^n e_{i,Pear}^2}{n-(p+1)}$ overdispersion scale parameter
 - p : # of explanatory variables used

Deviance Residuals

- $d_i =$
- $\text{sign}(y_i - \hat{\mu}_i) = 1$ if $y_i > \hat{\mu}_i$ and -1 if $y_i < \hat{\mu}_i$ and 0 if $y_i = \hat{\mu}_i$
- $D = \sum d_i^2$
- “Standardized” versions of Pearson and deviance residuals can be obtained by dividing by $\sqrt{1 - h_{ii}}$.
 - h_{ii} : Element from the hat matrix

Anscombe Residuals

- $e_i = \frac{f(y_i) - \hat{E}[f(y_i)]}{\sqrt{\widehat{Var}[f(y_i)]}}$
- $f(y_i)$: Function that makes the distribution of $f(y_i)$ as normal as possible

Concept Check #3

- Determine which statements are true.
- I. The Pearson chi-square statistic divided by degrees of freedom provides an estimate of the over-dispersion parameter.
- II. The scale parameter affects the model variance.
- III. The scale parameter affects model coefficient estimates.
- IV. The scale parameter affects the variance-covariance matrix.

4. Residuals Illustration

- *Values based off CAS Q.*
- n observations are exponentially distributed.

- $f(y_i) = \frac{e^{-\left(\frac{y_i}{\theta_i}\right)}}{\theta_i}, i = 1, 2, \dots, n$

- A GLM is fit to the data with the following model specification: $\ln(\theta_i) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{2i}^2 + \beta_4 x_{3i}$

- $\mathbf{y} = \begin{pmatrix} 14.8 \\ 137.6 \\ 0.4 \\ 38.3 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 1 & 1 & 2.7 & ? & 1 \\ 1 & 0 & 0.6 & ? & 1 \\ 1 & 1 & 2.9 & ? & 1 \\ 1 & 1 & 3.0 & ? & 0 \end{pmatrix}, \hat{\boldsymbol{\beta}} = \begin{pmatrix} 2.99 \\ -0.27 \\ -0.67 \\ 0.16 \\ 0.91 \end{pmatrix}$

- Calculate the Pearson and deviance residual for the first observation.

4. Residuals Illustration 2

- Pearson residuals $e_{i,Pearson} = \frac{y_i - \hat{\mu}_i}{\sqrt{Var(y_i)}}$
- $d_i = \text{sign}(y_i - \hat{\mu}_i)\sqrt{D}$
- $Var(y) = \phi v(\mu)$
- Exponential: $D = 2 \sum \left(\frac{y_i}{\hat{\mu}_i} + \ln \frac{\hat{\mu}_i}{y_i} - 1 \right)$

Tweedie Distribution

- The Tweedie distribution (aka. Compound Poisson-Gamma) is a commonly used mixture because it has a
- It is a Poisson sum of gamma random variables, and is used to model **aggregate loss** in insurance.
- $S_N = y_1 + y_2 + \cdots y_N$
 - S_N : Tweedie distribution value (aggregate loss)
 - N : Poisson random variable (# of claims)
 - y_i : Gamma distribution (claim amount)

Tweedie Distribution FYI

- In the Tweedie distribution,
- $P(S_N = 0) = P(N = 0) = e^{-\lambda}$
- $P(S_N \leq y) = e^{-\lambda} + \sum_{n=1}^{\infty} P(N = n)P(S_n \leq y)$
- $f_S(y) = \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \frac{\gamma^{n\alpha}}{\Gamma(n\alpha)} y^{n\alpha-1} e^{-y\gamma}$

Variance Function Summary

- $Var(y) = \varphi * \begin{cases} 1, \text{Normal } (p = 0) \\ \mu, \text{Poisson } (p = 1) \\ \mu^p, \text{Tweedie } (1 < p < 2) \\ \mu^2, \text{Gamma } (p = 2) \\ \mu^3, \text{Inverse Gaussian } (p = 3) \end{cases}$

Likelihood Ratio Test

- If H_0 is a nested (subset) model of H_a using a subset of explanatory variables with q parameters removed, then the likelihood ratio test statistic is:
- $LRT =$
 - H_0 : has $p + 1 - q$ total parameters
 - H_a : has $p + 1$ total parameters
- $D_0 = 2(l_{SAT} - l_0)$ and $D_a = 2(l_{SAT} - l_a)$

LRT Visual

H_0 : smaller model

ie. $g(y) = b_0 + b_1x_1 + b_3x_3$

H_a : larger model including all in H_0

ie. $g(y) = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$

LRT Q

Is larger l_a
or smaller D_a
"worth" the
extra q parameters

Model Selection

- Model selection should not be based on the largest likelihood, smallest deviance.
- Penalized metrics such as AIC/BIC are better.
- FYI: Formulas not tested
- $AIC = -2l + 2K$
- $BIC = -2l + K \ln n$
 - l : Model log-likelihood
 - K : # fitted parameters (including intercept, scale parameter if estimated)
 - n : # of observations in data

Chi-Square Table

- The chi-square test is always one-sided, so use the given significance levels directly.

df	$\chi^2_{0.100}$	$\chi^2_{0.050}$	$\chi^2_{0.025}$	$\chi^2_{0.010}$	$\chi^2_{0.005}$
1	2.706	3.841	5.024	6.635	7.879
2	4.605	5.991	7.378	9.210	10.597
3	6.251	7.815	9.348	11.345	12.838
4	7.779	9.488	11.143	13.277	14.860
5	9.236	11.070	12.833	15.086	16.750
6	10.645	12.592	14.449	16.812	18.548
7	12.017	14.067	16.013	18.475	20.278
8	13.362	15.507	17.535	20.090	21.955
9	14.684	16.919	19.023	21.666	23.589

5. LRT Illustration (CAS)

- A study was commissioned on the effect of type of fertilizer and type of seed on crop yield. There are three types of fertilizers, and four types of seed included in the study. Two separate GLMs with log link functions were fit to the dataset.
- Using type of fertilizer and type of seed without an interaction term, the deviance is 43.
- Using type of fertilizer and type of seed, and all interaction terms between those two main effect variables, the deviance is 30.
- H_0 : The effect of type of fertilizer is independent of type of seed on crop yield.
- H_a : The effect of type of fertilizer is not independent of type of seed on crop yield.
- Calculate the smallest significance level at which you reject H_0 .

5. LRT Illustration

- $LRT = 2(l_a - l_0) = D_0 - D_a$ is approximately χ_q^2

Example 6

- Several versions of an Inverse Gaussian with canonical link GLM are run and the following log-likelihoods are obtained:
- Intercept-only model: -1000
- Intercept and predictors X_1 : -997
- Intercept and predictors X_1, X_2, X_3 : -996
- A. $H_0: \beta_1 = \beta_2 = \beta_3 = 0$ is tested using the likelihood ratio test. What is the lowest significance level in the Tables where H_0 rejected?
- B. Repeat for $H_0: \beta_2 = \beta_3 = 0$
- C. What is the the pseudo- R^2 of the 1 predictor model if the pseudo- R^2 for the 3-predictor model is 75%?

Example 6 Sol

- $LRT = 2(l_a - l_0) = D_0 - D_a$ is approximately χ_q^2
- $\text{Pseudo-}R^2 = \frac{l - l_{Null}}{l_{SAT} - l_{Null}}$

Concept Check #4

- Determine which statements are true.
- I. The gamma distribution is suitable for modeling claim severities with a constant coefficient of variation.
- II. If the LRT rejects the null hypothesis of an intercept-only GLM in favor of one with 4 parameters, we can conclude the 4 parameters each should have a non-zero coefficient.
- III. GLMs assume predictors are not random (deterministic).
- IV. The LRT can only compare nested models.

HW Notes

- In some old CAS/CA Qs, they will use
- $D = -2l$
- That is not true, but using this to solve for model rankings will give you the correct answer, as $2(l_a - l_0) = D_0 - D_a$ always holds.
 - l_{SAT} cancels out
- The chi-square GOF test (from Statistics VEE) is no longer be on the SRM syllabus, but CA HW kept a few Qs on it that you can ignore.

Summary Check

- GLM model assumptions, how parameter estimation works, how to predict
- GLM canonical link functions, mean-variance relationships for distributions
- Linear exponential family decomposition
- Formulas for deviance, max-scaled R^2 , pseudo- R^2 , Pearson residual, deviance residual, overdispersion parameter, likelihood ratio test